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Diploma in Engg. (Third Semester) Examination, Nov.-Dec. 2015

(Old Course)

(Mech. Branch)

BASIC ELECTRICAL & ELECTRONICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *All questions are compulsory unless mentioned otherwise.*

1. Choose the correct answer : 5×2=10

(i) Electrical appliances are connected in parallel

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because it :

(a) is a simple circuit

(b) draws less current

(c) result in reduce power loss

(d) makes the operation of appliances independent of each other

(ii) A d.c. motor is used to .....

(a) generate power

(b) change mechanical energy to electrical energy

(c) change electrical energy to mechanical energy

(d) increase energy put it in it

(iii) A transformer is an efficient device because it :

(a) is a static device

(b) uses inductive coupling

(c) uses capacitive coupling

(d) uses electric coupling

(iv) If doping level in a crystal diode is increased with of the depletion layer .....

(a) remains the same

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(b) is increased

(c) is decreased

(d) none of above

(v) An a meter is ..... instrument.

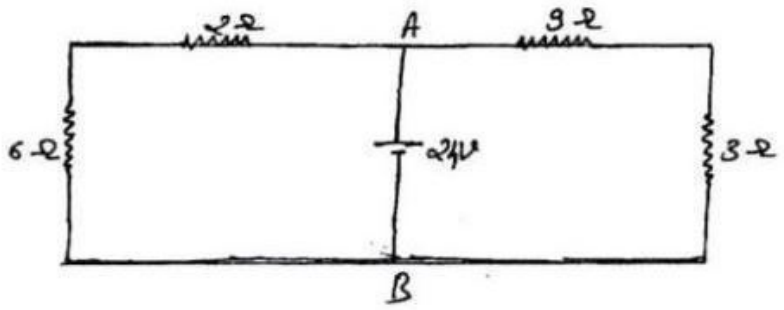
(a) an indicating

(b) an integrating

(c) a recording

(d) none of above

2. (a) In the circuit shown below, find the voltage and current in each element. 5



(b) State and explain KVL and KCL. 5

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3. A 230 V, 50 Hz a.c. supply is applied to a coil of 0.06 H inductance and 2.5 Ω resistance connected in series with a 0.68 μF capacitor.

Calculate :

(i) impedence

(ii) current

(iii) phase angle between current and voltage

(iv) power factor; and

(v) power consumed 10

4. Write down the types of Single Phase Motors? Explain Capacitor start capacitor-run motor and shaded pole motor with its working and constructional details. 10

5. Explain the voltage and currents in balanced : 10

(a) Δ-- delta connection. And investigates its characteristic also.

Or

(b) A balanced 3-phase load consist of three coils, each of resistance 6 Ω and inductive reactance of 8 Ω. Determine the line current and power absorbed when the coils are :

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(i) Star connected

(ii) Delta connected across 400 V 3 phase supply

6. Explain the construction and working of Energy meter. 10

Or

Explain Dynamometer type wattmeter with its construction? Derive the deflecting torque of it?

7. Explain the principle of working of d.c. generator? Also draw wave diagram. 10

Or

A 4 pole d.c. shunt generator with a wave wound armature has to supply a load of 500 lamps each of 100 W at 250 V. Allowing 10 V for the voltage drop in the connecting leads between the generator and the load and drop of 1 V per brush. Calculate the speed at which the generator should be driven. The magnetic flux per pole is 30 mwb and the armature and shunt field resistances are respectively 0·05 Ω and 65 Ω. The number of armature conductor is 390.

8. Explain the working principle of a transformer with its circuit diagram. 10

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Or

Derive e.m.f. equation of a D.C. generator.

9. Explain the working of full wave bridge rectifier with the help of circuit diagram and wave diagram. 10

Or

Explain :

(i) Amplifier action of transistor

(ii) SCR and its application

10. Explain short circuit test of a Transformer with its circuit diagram.

Or

Write short notes on : (any **two**) 10

(i) Application of universal motor

(ii) Transformation ratio

(iii) Efficiency of transformer

(iv) Making self starting single phase induction motor

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